

Mask-Free Vision-Language Pretraining for Cross-Domain Few-Shot CT and MRI Segmentation

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Abstract—Pixel-level segmentation masks remain scarce for most medical imaging sources, while paired image-text archives such as radiology case reports have stayed comparatively abundant yet carry no masks at all. This paper has studied whether a lightweight, text-conditioned U-Net can be pretrained purely on unpaired image-text pairs for a set of anatomical categories and then transferred to a completely unseen target organ using only a handful of real segmentation examples. We have pretrained a shared image-text encoder via contrastive learning on MedPix 2.0, with the target organ’s category withheld entirely, then adapted a segmentation decoder on a different organ and measured a 0/10/20/50-shot data-efficiency curve across four dataset and organ combinations: two organs on SLAKE and two independent external braintumor datasets. A retrieval-based diagnostic has localized the zero-shot weakness to the contrastive pretraining stage itself rather than to the downstream segmentation architecture. Across a broad ablation spanning loss functions, attention, optimizers, text-conditioning design, and episodic meta-learning, only an ImageNet pretrained backbone and pseudo-label self-training have produced confirmed gains, closing two of four combinations to within or above 80% of their architecture’s own oracle ceiling; every meta-learning extension leaves the zero-shot gap unchanged. This convergent negative evidence is the paper’s central finding: the residual gap is set upstream, during pretraining, rather than by anything downstream of it.

Index Terms—medical image segmentation, few-shot learning, cross-domain transfer, text-conditioned segmentation, contrastive pretraining, meta-learning, U-Net, data efficiency

I. INTRODUCTION

Deep learning segmentation models require dense, pixel-level annotations, the single most expensive resource in medical imaging: they demand expert radiologist time and rarely transfer cleanly across institutions or organs. Medical imaging archives with paired free-text descriptions, such as radiology reports, remain comparatively plentiful but carry no segmentation masks. This raises a practical question: can a model pretrained on abundant, mask-free image-text data be adapted to segment a completely novel organ using only a handful of real labeled examples, and how many examples are actually required before performance becomes clinically useful? We study this under a deliberately strict protocol: we pretrain a shared image-text encoder via contrastive learning on MedPix 2.0 [1], with every image belonging to the eventual target organ’s category withheld entirely from pretraining, then adapt a segmentation decoder using only real masks from a different organ, and evaluate on the withheld target organ at 0, 10, 20,

and 50 real fine-tuning examples, a data-efficiency curve rather than a single headline number. This leave-one-category-out design directly measures both zero-shot cross-organ transfer and the marginal value of each additional labeled example, which a single train/test split cannot distinguish.

A. Contributions

This study addresses the research gap identified in Section II with the following contributions.

- **A mask-free pretrain, adapt, and fine-tune pipeline** for cross-organ few-shot segmentation using only contrastive image-text pretraining plus a small amount of real target organ supervision, reported as a full 0/10/20/50-shot curve; the adopted recipe closes two of four combinations to above 80% of the architecture’s own oracle ceiling.
- **Data-efficiency results across four combinations** (SLAKE head, SLAKE abdomen, FigShare, LGG MRI), each benchmarked against its own oracle ceiling and, where available, the literature state of the art; our LGG result reaches 82.4% of published state of the art after self-training, within two points of human inter-rater agreement.
- **A retrieval-based diagnostic**, measured entirely within the pretraining corpus, that isolates the zero-shot generalization gap to contrastive pretraining: held-out image to text Recall@1 collapses to 0.11–2.19%, near chance, while in-distribution Recall@1 reaches 80–142× chance.
- **A broad ablation study** spanning loss functions, attention, optimizers, text-conditioning design, and episodic meta-learning; only an ImageNet pretrained backbone and pseudo-label self-training produce confirmed gains, and no intervention closes the zero-shot gap itself.

II. LITERATURE REVIEW

Text-guided medical segmentation typically conditions on free-text or structured prompts via cross-attention or feature-wise modulation, via a vision transformer with language input at multiple decoder scales [8] or via text-descriptor guidance augmenting the Segment Anything Model [9], both reporting Dice gains over their respective baselines; the latter finds structured shape and location descriptors can matter more than the fusion mechanism for organs unseen during training. Our text ablation and richer text experiments (Section IV-F)

directly test this claim and find it does not replicate in our setting, which we attribute to our text encoder’s grounding being learned at a much smaller scale.

The closest prior work in dataset pairing is Xu *et al.* [24], who pretrain a unified vision-language transformer on Radio-Graphy Captions, 18,434 image and caption pairs from the same MedPix source, and transfer to SLAKE among other benchmarks, reporting accuracy on visual question answering, generation quality on report writing, and recall on image to text retrieval, all under full in-domain supervision with no held-out category. Ours is pixel-level segmentation specifically, under an explicit leave-one-category-out protocol with the target category entirely absent from pretraining.

Closer to our mechanism is CT-GLIP [25], pretrained on 44,011 organ-level CT-report pairs from 17,702 patients covering 104 organs, which reports a 3.2% DSC gain over CLIP-style global alignment on zero-shot tumor segmentation, showing the same mechanism succeeds at roughly $20\times$ our corpus scale with organ-level grounding: external evidence for our corpus-scale conclusion (Section V) rather than a threat to it.

Our problem setting otherwise sits within cross-domain few-shot medical segmentation, addressed via discriminative prototype embeddings [10], superpixel-based pseudo-masking [11], episodic meta-learning over support and query pairs [12], MedCon’s unified contrastive pretraining across diverse domain shifts without per-shift tailoring [26] (a fixed label space, not category exclusion), and Zhu *et al.*’s domain-robust feature selection (DSM), related to Squeeze and Excitation [21], which motivates one of our core ablations (Section IV-F) [13]. We report a further post-hoc episodic meta-learning exploration, also in Section IV-F, adapted from the domain generalization formulation of Li *et al.* [23] rather than [12]’s support/query adaptation, since our setting has zero labeled target-category examples at any pretraining stage, not merely few. A large-scale loss-function survey [14] and boundary-distance weighting for small lesions [15] motivate our Kervadec-style boundary loss [16] ablation (Section IV-F), where we report a carefully verified null result.

A. Research Gap

Existing text-conditioned and cross-domain few-shot segmentation methods either assume the target category is represented, even sparsely, in the pretraining corpus, or evaluate transfer only on the downstream segmentation architecture without isolating whether a failure originates upstream, during pretraining, or downstream, in the decoder. Prior grounded vision-language-to-segmentation transfer work, including CT-GLIP [25], operates at large pretraining scale without an explicit held-out-category protocol; we isolate the small-corpus, strict-exclusion regime specifically, pairing MedPix derived contrastive pretraining with a segmentation transfer target under a leave-one-category-out discipline where the target category has zero exposure at any pretraining stage. None of the work reviewed above combines a retrieval only diagnostic with a systematic ablation to localize where a cross-

organ transfer failure actually originates. This is the gap this paper has addressed, on two organs from SLAKE [2] (head, abdomen) and, as independent external validation, on FigShare Brain Tumor [3] and LGG MRI Segmentation [4], using the identical Phase A checkpoint and Phase B adaptation data as the SLAKE head result, so all four curves are directly comparable.

III. PROPOSED METHODOLOGY

This paper has proposed a method for cross-organ few-shot medical image segmentation from mask-free image-text pretraining. Fig. 1 summarizes (a) the model architecture and (b) the three-phase training pipeline detailed in this section.

A. Architecture

Fig. 1(a) shows the four components end to end: an image encoder, a text encoder, a bottleneck cross-attention fusion module, and a U-Net-style decoder with skip connections, with per-stage resolutions and channel counts as labeled. The image encoder is an ImageNet pretrained ResNet-18 [5], frozen except its final residual block, in place of a from-scratch encoder that has no visual prior for an organ it has never seen (empirically motivated, Section IV-F). The text encoder is PubMedBERT base [6], frozen except its final transformer layer; PubMedBERT’s $\sim 110\text{M}$ parameters dominate the total parameter count, so we report the trainable count (11–17M depending on configuration) separately as the figure relevant to the model’s lightweight design intent. We additionally implement and separately evaluate FiLM [7] conditioning of every decoder skip connection, and a fully text-free ablation bypassing fusion entirely (Section IV-F).

B. Three-Phase Training Pipeline

Phase A, contrastive pretraining. The shared image-text encoders are trained via InfoNCE-style contrastive image-text alignment on MedPix 2.0, with every image belonging to the target organ’s *Location Category* excluded from the training pool entirely. No masks are used or needed at this stage.

Phase B, decoder adaptation. Using the Phase A checkpoint, a segmentation decoder is trained on real masks from a different organ than the eventual target, never the same organ being evaluated: SLAKE head adapts on SLAKE abdomen, SLAKE abdomen reverses this and adapts on SLAKE head, and FigShare and LGG, which contain only brain and head data, both adapt on SLAKE abdomen, the only non-brain organ available, using the same head-excluded Phase A checkpoint as the SLAKE head result so all three head/brain curves are directly comparable. SLAKE abdomen’s 0-shot result in Table II therefore had no exposure to abdomen masks prior to Phase C.

Phase C, few-shot fine-tuning. The Phase B model is evaluated zero-shot on the target organ’s held-out evaluation set, then incrementally fine-tuned on cumulative draws of 10, 20, and 50 real target organ examples; the 50-shot model has seen the 10 and 20-shot examples plus 40 new ones, not an independent draw. A single optimizer instance is persisted

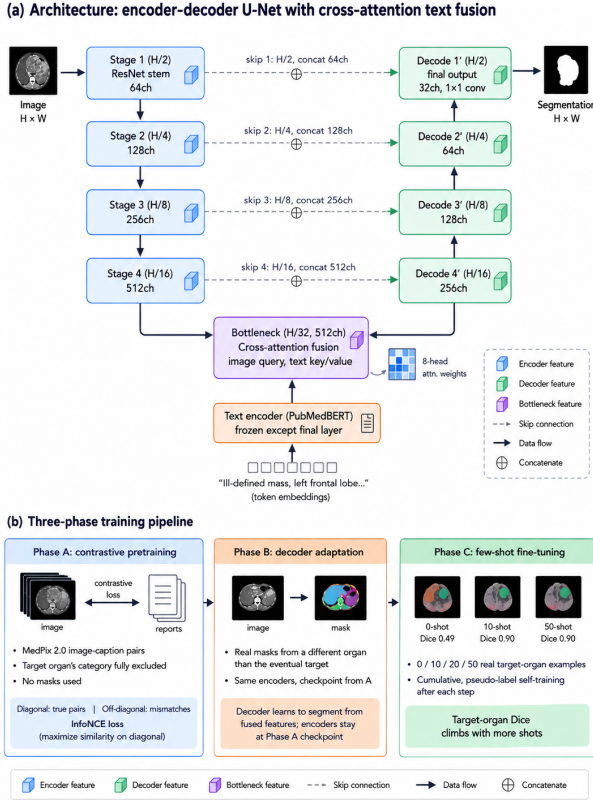


Fig. 1. Overview of the proposed method. (a) ResNet-18 image encoder, PubMedBERT text encoder, bottleneck cross-attention, and U-Net decoder. (b) Three-phase training: contrastive pretraining, decoder adaptation, and cumulative few-shot fine-tuning on the withheld target organ.

and its learning rate decayed across increments, rather than reinitialized at each step, which we found necessary to avoid a transient instability between the 10 and 20-shot increments in early experiments.

IV. EXPERIMENTAL RESULTS

A. Datasets

Table I summarizes all four datasets. MedPix 2.0 [1] supplies the leave-one-category-out exclusion filter via its five coarse *Location Category* labels and carries no segmentation masks. SLAKE [2] is the primary segmentation target; we construct our own image-level train/eval split, since SLAKE’s released split is defined at the question-answer pair level and is not image disjoint, and binarize all annotated structures to a single foreground class. FigShare Brain Tumor [3] and LGG MRI Segmentation [4] serve as external, brain and head only validation, both split at the patient level to avoid leakage; LGG is CC-BY-NC-SA-4.0 (noncommercial) licensed, notably stricter than the other three datasets, which we flag explicitly since a common third-party mirror lists it incorrectly as plain CC-BY-4.0. All three segmentation-target datasets were checked for image-level overlap against MedPix 2.0 and against each other via perceptual hashing followed by pixel-correlation confirmation; no confirmed duplicates were found.

B. Evaluation Protocol

Oracle ceiling. For each combination we additionally train the same architecture on all available target organ data, using the Phase A checkpoint that did not exclude that organ, establishing each combination’s own achievable ceiling, the denominator for the percent of oracle reached metric.

Retrieval diagnostic. To isolate whether weak few-shot results originate from contrastive pretraining or the downstream segmentation task, we measure image to text Recall@1 and Recall@5 entirely within MedPix 2.0, separately on held-out target organ images and a size-matched in-distribution sample; a large gap indicates the shared encoder has failed to build a meaningful representation of an organ it has never seen. The same diagnostic is reused, unmodified, for the episodic Phase A meta-pretraining variant in Section IV-F.

Self-training. After each real-shot Phase C step, the model pseudo-labels up to 30 further unlabeled pool images, filters out degenerate pseudo-masks with foreground fraction outside [0.001, 0.5], and refines on a mixture of real (weight 1.0) and pseudo-labeled (weight 0.3) examples.

Loss function. The base segmentation loss is $\mathcal{L}_{\text{seg}} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}}$; Section IV-F describes eight further interventions evaluated on top of this baseline.

C. Evaluation Metrics

We report four metrics. **Dice** is the standard region-overlap coefficient between predicted and ground-truth foreground masks, unitless, 0 to 1, higher is better; reported as a raw coefficient in Tables II and VI and separately as $\text{Dice} \times 100$ (%) in Table V where noted. **HD95** is the 95th-percentile symmetric boundary distance in pixels between predicted and ground-truth foreground regions, lower is better, and is less sensitive to outlier pixels than the standard Hausdorff distance. **Recall@k** (**R@k**) is the fraction of retrieval-diagnostic query images for which the paired caption appears among the top- k retrieved captions by cosine similarity, reported as a percentage, higher is better.

Percent of Oracle Reached. The few-shot Dice score expressed as a percentage of the same architecture’s oracle ceiling Dice, normalizing for each dataset’s inherent difficulty; always reported as a percentage (%).

$$\% \text{ of Oracle} = 100 \times \frac{\text{Dice}_{\text{few-shot}}}{\text{Dice}_{\text{oracle}}} \quad (1)$$

D. Results

Table II reports Dice at each shot level for the adopted base recipe (ResNet-18 backbone, extended 20-epoch Phase C fine-tuning, text conditioning retained, base Dice+BCE loss) before self-training, over five random seeds per combination; standard deviations ranged approximately 0.01–0.08 depending on shot level and combination, largest at low shot counts as expected. Self-training is applied as a further step on top of this curve at every shot level (Section IV-F), so the curve here remains directly readable on its own.

All four combinations show the same qualitative pattern: weak zero-shot transfer, a clear monotonic improvement from

TABLE I
DATASET SUMMARY

Dataset	Task	Samples	Train/pool	Eval/test	Description
MedPix 2.0 [1]	Pretraining (retrieval)	2,050 images	all, target excluded	—	Radiology teaching-file archive, five Location Categories, no masks.
SLAKE [2]	Segmentation	310 usable images	218	92	VQA-derived masks, binarized; head (154) / abdomen (156).
FigShare [3]	Segmentation	3,064 slices, 233 patients	~70% patients	~30% patients	T1-weighted contrast-enhanced MRI, three tumor types.
LGG MRI [4]	Segmentation	1,373 non-empty slices, 110 patients	~70% patients	~30% patients	FLAIR MRI, TCGA-LGG, patient-level split.

TABLE II
DATA-EFFICIENCY CURVES (DICE), ADOPTED BASE RECIPE (BEFORE SELF-TRAINING), MEAN OVER 5 SEEDS

Dataset/Organ	0-shot	10-shot	20-shot	50-shot
SLAKE head	0.217	0.297	0.425	0.494
SLAKE abdomen	0.284	0.640	0.684	0.756
FigShare	0.109	0.164	0.242	0.349
LGG MRI	0.189	0.182	0.263	0.385

a handful of real examples, and continued gains through 50 shots with no saturation. SLAKE abdomen shows the steepest early gain (+0.356 from 0 to 10 shots), explained by three factors: $\sim 2.6\times$ larger segmentation targets (mean foreground fraction 8.22% vs. 3.18%), a more standardized 79% CT modality mix versus head’s 84% MRI, and a single homogeneous location category versus head’s three structurally distinct ones.

Fig. 2 gives Table II’s 50-shot column a visual, per-pixel counterpart: every combination’s best case is a tight match to ground truth by 50-shot, most cleanly for SLAKE abdomen, whose best case reaches Dice 0.93.

Table III reports the image to text retrieval diagnostic (Section IV-B), measured purely within MedPix 2.0.

TABLE III
IMAGE TO TEXT RETRIEVAL DIAGNOSTIC (MEDPIX 2.0 ONLY)

Excluded organ	Indist R@1	Held-out R@1	Indist R@5	Held-out R@5
Head	15.61%	0.11%*	47.06%	0.68%
Abdomen	24.76%	2.19%†	68.97%	4.39%

*Chance level for head ($n=884$) is $\approx 0.11\%$. †Chance level for abdomen ($n=319$) is $\approx 0.31\%$.

The model learns substantial, well above chance image-text alignment for organs it trains on ($80\text{--}142\times$ chance at Recall@1), but retrieval on the excluded organ sits at or barely above chance for both organs. This localizes the zero-shot weakness to the contrastive pretraining stage rather than to SLAKE specific domain shift or the segmentation decoder, and motivates the ablation study in Section IV-F.

Table IV reports HD95, computed over the full eval set for all four combinations using the identical production recipe as Table II (single seed; the same seed Dice values track the 5-seed means within 0.01–0.05 at every cell).

SLAKE abdomen shows the cleanest pattern, HD95 falling monotonically alongside Dice (67.6px to 25.9px). FigShare

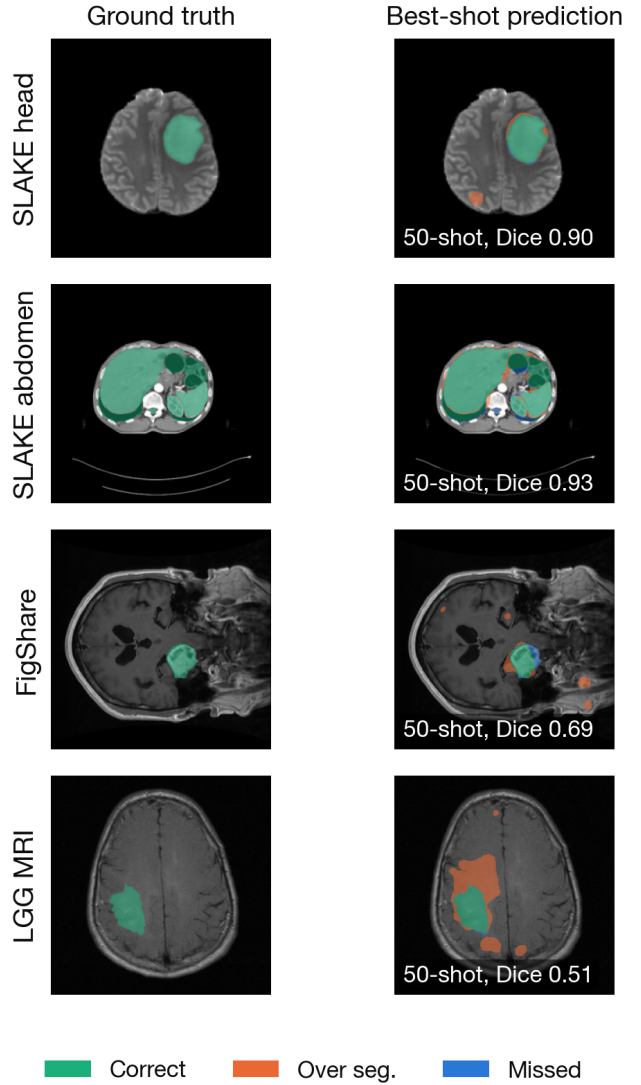


Fig. 2. Best case 50-shot prediction for all four dataset/organ combinations (single best performing example by Dice); overlay colors mark correct, over-segmented, and missed regions against the ground truth. Failure modes and the full shot-by-shot progression are already quantified via Tables II and IV and are not duplicated here.

TABLE IV
HD95 (PIXELS, LOWER IS BETTER), PRODUCTION RECIPE, SINGLE SEED,
FULL EVAL SET

Dataset/Organ	0-shot	10-shot	20-shot	50-shot
SLAKE head	62.33	54.15	49.46	51.26
SLAKE abdomen	67.59	50.76	40.44	25.87
FigShare	88.45	102.33	102.17	77.10
LGG MRI	84.80	86.64	78.62	66.56

is the case this metric earns its place on: Dice improves monotonically, but HD95 worsens from 0 to 20-shot (88.5px to 102.2px) before improving by 50-shot (77.1px), the opposite direction from Dice, evidence that early examples grow the predicted region before tightening its boundary to the true lesion edge.

E. Comparison

Table V reports each combination’s oracle ceiling, the fraction reached at every shot level before and after self-training, and, where a published benchmark exists, the literature state of the art; SLAKE has none (confirmed via literature search; all prior SLAKE work addresses VQA). Published state of the art and our figures are not directly comparable, since it uses full in-domain supervision with dedicated, heavily tuned architectures versus our deliberately harder transfer setting, but the comparison situates our architecture’s own ceiling relative to the field.

Self-training closes SLAKE abdomen and LGG MRI above the 80% target; SLAKE head and FigShare remain below it despite the ablation study in Section IV-F. Both external datasets’ oracle ceilings trail published state of the art by a wide margin, FigShare particularly so (67.0% vs. 87–97%), which we attribute to three design choices: zero data augmentation anywhere in Phase B/C/oracle training, one fixed oracle recipe with no per-dataset hyperparameter search, by design, and our binary foreground/background formulation, which discards FigShare’s three-way tumor-type structure that dedicated architectures exploit directly. LGG’s published state of the art (83.6%) is itself close to reported human inter-rater agreement (~84%), indicating that ceiling is bounded by annotation ambiguity, not model capacity.

F. Ablation Study

Following the diagnosis in Section IV-D, we have conducted a core ablation of ten interventions over five seeds, plus a post-hoc episodic meta-learning extension of three further interventions over one to four seeds given later time constraints, kept separate for that reason. Episodic meta-learning follows the MLDG-style domain generalization formulation of Li *et al.* [23]: training on episodes resembling the train/test mismatch faced at deployment, optimizing for gradient agreement across simulated source and target organs rather than empirical risk on pooled data. Each intervention is literature motivated and tested one variable at a time against the current best baseline; Table VI reports the mean Dice change at each shot level relative to that baseline.

Adopted (2). The ImageNet pretrained backbone quadrupled in-distribution retrieval while leaving held-out retrieval unchanged, and lifted Dice mainly at 20/50-shot (Table VI); its smaller, organ-dependent 0-shot gain reflects generic visual priors improving the decoder’s starting point, distinct from, and not evidence against, the retrieval diagnostic’s finding that held-out alignment itself does not improve. Self-training produced a consistent, non-negative effect at every shot level, closing SLAKE abdomen and LGG MRI above 80% of their own oracle ceiling.

Mixed (2). These move zero-shot performance without net benefit. SAM [19] gave a small 0-shot gain but a real regression by 10–50-shot, at $2\times$ compute. Episodic meta-training at Phase B with genuine three-organ episodes gave a mixed pattern against the production baseline, though it substantially mitigated damage from naive multi-source data pooling on the identical data (vs. a -0.087 0-shot regression for the non-episodic pooled control).

Null (9). Focal Tversky loss, attention gates, richer text prompts, multi-scale FiLM, text-conditioning removal, boundary loss, a domain-robust feature gate, and both remaining episodic meta-learning variants produced no confirmed positive effect distinguishable from noise, and none is adopted. No intervention tested closes the zero-shot gap: combined with the retrieval diagnostic (Section IV-D), this is convergent evidence that the residual gap for SLAKE head and FigShare is set upstream, during Phase A pretraining, surviving even episodic meta-learning, the field’s standard remedy for exactly this problem.

V. CONCLUSION AND FUTURE WORK

This paper has presented a text-conditioned, lightweight U-Net for cross-domain few-shot medical image segmentation, pretrained via contrastive image-text alignment with a target organ entirely withheld, evaluated as a 0/10/20/50-shot data-efficiency curve across four dataset and organ combinations. Two combinations, SLAKE abdomen and LGG MRI, reach above 80% of their own oracle ceiling after adopting an ImageNet pretrained backbone and self-training; SLAKE head and FigShare do not, despite a broad ablation finding no fix. A retrieval-based diagnostic, corroborated by every intervention tested, including episodic meta-learning explicitly designed to reach further upstream, localizes this residual gap to the contrastive pretraining stage itself.

Corpus scale and category diversity at Phase A are the most direct remaining candidate bottlenecks: MedPix 2.0’s five Location Categories are thin by meta-learning literature standards, the most plausible reason even episodic meta-learning could not close the gap. We have not directly manipulated corpus scale in this study, so this remains a candidate diagnosis; CT-GLIP’s zero-shot tumor segmentation success at roughly $20\times$ our corpus scale [25] is external evidence consistent with it. A scale ablation at Phase A is the top priority for future work, alongside tumor-type-aware modeling for FigShare and extending HD95 to the full five-seed protocol.

TABLE V

ORACLE CEILING, PERCENT REACHED AT EVERY SHOT LEVEL, AND PUBLISHED STATE OF THE ART. ALL ENTRIES ARE DICE EXPRESSED AS A PERCENTAGE (DICE $\times$ 100), EXCEPT THE INTERVENTION COUNT COLUMNS WHICH ARE UNITLESS SHOT COUNTS.

Dataset/Organ	Oracle (%) Dice $\times$ 100	0	10	20	50 (base)	50 (+ST)	State of the art (%) full-data [‡]
		% of oracle reached					
SLAKE head	67.5%	32.1%	44.0%	63.0%	73.2%	73.9%	n/a
SLAKE abdomen	85.6%	33.2%	74.8%	79.9%	88.3%	90.0%	n/a
FigShare	67.0%	16.3%	24.5%	36.1%	52.1%	51.9%	87–97% [22]
LGG MRI	48.5%	39.0%	37.5%	54.2%	79.4%	82.4%	83.6% [4]

[‡]Published state of the art is trained on the full labeled training set for that dataset, not a shot-limited regime, and is not directly comparable to the 0/10/20/50-shot columns to its left.

TABLE VI

ABLATION SUMMARY WITH MEASURED DICE CHANGE (Δ) RELATIVE TO THE CURRENT BEST BASELINE AT EACH SHOT LEVEL, MEAN ACROSS ORGANS WHERE APPLICABLE. Δ VALUES ARE RAW DICE POINTS (0 TO 1 SCALE, NOT A PERCENTAGE), EXCEPT THE ROW MARKED *, WHICH IS IN PERCENTAGE POINTS OF RECALL@1.

Intervention	$\Delta 0$	$\Delta 10$	$\Delta 20$	$\Delta 50$	Status
ImageNet pretrained backbone [5]	+0.057	+0.005	+0.080	+0.103	Adopted
Self-training [20]	—	+0.027	+0.030	+0.008	Adopted
SAM optimizer [19]	+0.007	−0.067	−0.084	−0.123	Mixed
Episodic meta, 3-source Phase B [23]	+0.049	−0.034	+0.014	—	Mixed
Focal Tversky loss [17]	−0.058	−0.110	−0.188	−0.258	Null
Attention gates [18]	−0.030	−0.053	−0.102	−0.101	Null
Richer text prompts [9]	+0.011	+0.001	+0.003	+0.000	Null
Multi-scale FiLM [7], [8]	+0.015	+0.020	+0.003	−0.003	Null
Text-conditioning removal	+0.006	+0.004	+0.000	−0.006	Null
Boundary loss [14], [16]	−0.009	+0.006	−0.009	−0.005	Null
Domain-robust feature gate [13]	−0.018	−0.024	−0.012	−0.007	Null
Episodic meta-pretrain, Phase A [23]	−0.055*	n/a	n/a	n/a	Null
Episodic meta, single-source Phase B [23]	+0.012	−0.006	n/a	n/a	Null

*This row reports Δ held-out Recall@1 in percentage points, not Δ Dice, since Phase A meta-pretraining is evaluated on the retrieval diagnostic (Section IV-B), not on segmentation; the value shown is the mean of the two tested organs (head −0.11pp, abdomen 0.00pp).

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